

π _RL · arXiv 2510.25889 · Oct 5

Empiricist

11+3 · seat weight 4 · HIGH

Smoke-feasible grid + tattoo numbers

Defer full online RL

Kanta not seated — all five packs built

Spine: unlock PPO on flow VLAs; BEHAVIOR still prefers RFT

Chen / Liu / Zhang et al. · RLinf · ICML-style (TENTATIVE)

Club spine · Oct 5

Unlock PPO on flow VLAs · BEHAVIOR still prefers RFT

BLOCKED

Flow VLAs

intractable $\log \pi$
+ deterministic ODE
→ no PG / no explore

π_{RL} OPENS

Flow-Noise / SDE

tractable Gaussian
+ PPO/GAE on $\sim 300M$
expert (VLM frozen)

FAST SIM

LIBERO / ManiSkill

+27-31 pp ID avg
few-shot+RL > full SFT
partial env OOD

SLOW SIM

BEHAVIOR / OmniG

Comet cites π_{RL}
then uses RFT
sample-efficiency gate

When to use π_{RL}

- Parallel sparse-success sim available
- OpenPI π_0 / $\pi_{0.5}$ (or GR00T) already SFT'd
- Want ID specialists beyond demos
- Can afford 8×H100-class + 64-320 envs
- Real deploy = sim RL → ODE eval (not real RL)

When to stay on RFT

- Slow household sims (BEHAVIOR)
- Heterogeneous GPU / RT-core split pain
- Need coverage flywheel, not on-policy
- Comet path: reject → SFT (Q 0.224 → 0.345)
- 2026: compose RFT then short π_{RL} burst

Tattoo numbers · Empiricist must memorize

π_0 SFT → Noise

51.1 → 80.3

+29.2 pp ID avg

$\pi_{0.5}$ SFT → SDE

55.6 → 86.6

+31.0 pp ID avg

LIBERO Long

43.9 → 94.0

$\pi_{0.5}$ Flow-Noise

Few-shot+RL

98.3% > 96.9%

beats full $\pi_{0.5}$ SFT

Real2Sim2Real

SFT fail → 40%

3DGS · thin n

Extras

PPO > GRPO ~6pp

GR00T 52.5→89.9

Also tattoo

ManiSkill $\pi_{0.5}$ 40.1→90.9 IND · OOD ~26→49-53 · CALVIN Len-5 61.3→87.0 · ML45 novel tasks: NO stable gain

Honesty: 'beats full SFT' omits interactive sample cost · OOD = env/visual, not open-world discovery

Table 1 · ID average success % · tattoo these

Four suites averaged · Δ vs few-shot SFT

Model	Method	LIBERO	ManiSkill	MetaWorld	CALVIN	Avg	Δ
π_0	SFT	57.6	38.4	50.8	57.5	51.1	—
π_0	Flow-SDE	96.1	78.8	78.1	61.7	78.7	+27.6
π_0	Flow-Noise	97.6	77.8	85.8	59.9	80.3	+29.2
$\pi_{0.5}$	SFT	77.1	40.1	43.8	61.3	55.6	—
$\pi_{0.5}$	Flow-SDE	97.9	90.9	70.7	87.0	86.6	+31.0
$\pi_{0.5}$	Flow-Noise	98.3	89.7	66.1	84.5	84.7	+29.1

π_0 SFT→Noise

51.1 → 80.3 (+29.2)

$\pi_{0.5}$ SFT→SDE

55.6 → 86.6 (+31.0)

Best avg

$\pi_{0.5}$ Flow-SDE 86.6

Ruthless smoke priority · defer full online RL

Empiricist matrix — P0 Y / P1 Partial / Defer N

P P0	Smoke Y	Experiment Released ckpt eval	What OpenPI / RLinf weights on 1 suite vs SFT
P0	Y	Eval-only Kxa grid	$a \in \{0.2, 0.5, 0.8\} \cdot K \in \{1, 2, 4, 8\} \cdot$ no PPO
P0	Y	Unit-test Flow-SDE	Toy dim · ODE vs SDE marginal · hybrid
P1	Partial	Offline AW-CFM	Advantage-weighted CFM on logged rollouts
P1	Partial	Critic warm-up	V_{vlm} vs V_{expert} on fixed buffer
P1	Partial	Tiny PPO ≤ 8 envs	$\leq 20\text{--}40$ updates · not Tab.1 reproduction
Defer	N	Tab.1 multi-bench	$8 \times H100 \cdot 4$ suites · hundreds of epochs
Defer	N	ManiSkill 4k / Real / BEHAVIOR	Weeks + hardware — out of quiet scope

Hypotheses · H1 / H2 / H4 / H5

Feasible track — falsifiable without full Tab.1

H1

mid noise $a \approx 0.5$

maximizes usable train-mode SR
low $a \rightarrow$ high clip; high $a \rightarrow$ collapse

H2

$K \in \{4,5\}$

operating point; diminishing returns
by $K \approx 8$ · $K=1$ breaks ODE-SDE

H4

freeze VLM + expert

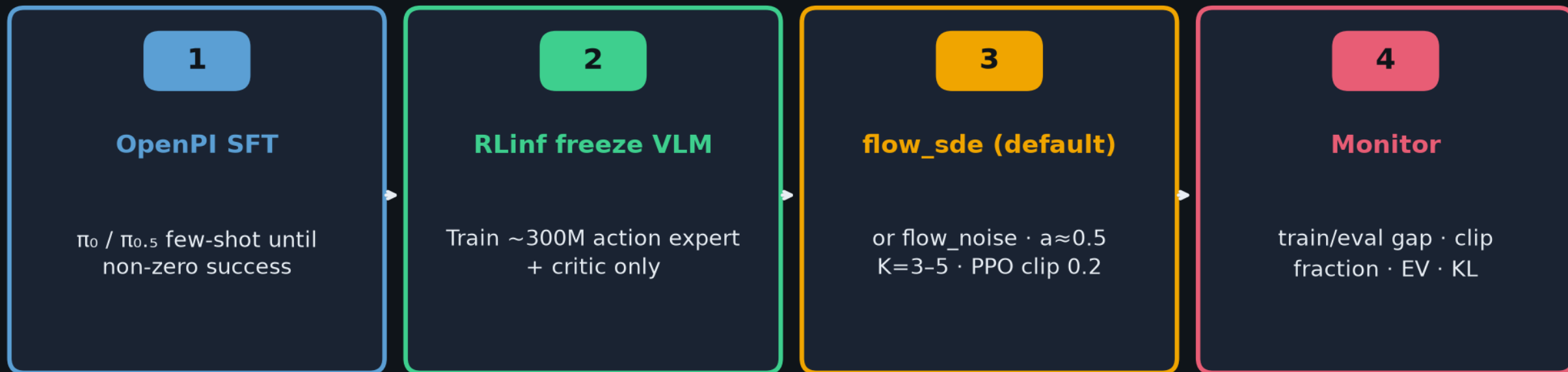
ID refinement — NOT novel-task
invention (ML45 spirit)

H5

mid H + replan

> cargo-cult $H=50$ during RL
EV dies on long chunks

Reproduce outline · OpenPI → RLinf → flow_*



One-liner

parallel sparse-success sim → near-saturated specialists;

else stay on RFT

Never claim Tab.1 numbers from Partial smokes

GPU honesty · Empiricist / Stakeholder

Workload	VRAM	Wall	Smoke
Unit-test Flow-SDE	<2 GB	minutes	Y
Eval-only Kxa grid	12-24 GB	hours	Y
Critic fit	8-20 GB	hours	Y
Tiny PPO ≤ 8 envs	24-48 GB	1-3 days	Partial
Paper LIBERO-style	8×H100	days-weeks	N
BEHAVIOR online RL	multi extreme	out of scope	N

Control note · not classical planner

Flow → MDP

Stochastic via Flow-SDE
or Flow-Noise train path
Deploy = ODE Euler

Hybrid sampler

1 SDE step / env step
rest ODE → $\sim 2\times$ faster
vs full two-layer

PPO on expert

$\sim 300\text{M}$ action expert
VLM frozen default
chunk-level reward

Chunk length H = credit-assignment knob

Paper: π_0 LIBERO $H=50$ · $\pi_{0.5}$ $H=10$ · ManiSkill $H=8$ · CALVIN/MW $H=5$

Larger chunks $\uparrow$ SFT baseline but $\downarrow$ explained variance / RL ceiling

Empiricist: prefer Comet-like mid horizons + receding H' over cargo-cult $H=50$

Flow-Noise vs Flow-SDE · two ways to unlock PPO

Both train stochastic · both deploy as deterministic ODE (noise discarded / SDE off)

Flow-Noise

ReinFlow-inspired · one-layer MDP

- ▶ Learnable $\sigma_{\theta'}(A^{\tau}, o) \rightarrow$ Gaussian denoise step
- ▶ Exact joint log-prob of denoise chain (eq. 5)
- ▶ `joint_logprob=True` · entropy ~ 0.005
- ▶ Noise net discarded after fine-tune
- ▶ Often wins MetaWorld (π_0 85.8 vs 78.1)
- ▶ Cost: full denoise recompute for likelihood

Flow-SDE

Flow-GRPO / DPPO · two-layer + hybrid

- ▶ ODE $\rightarrow$ equivalent SDE (Song PF-ODE)
- ▶ Noise $\propto \sqrt{\tau/(1-\tau)}$; Gaussian step likelihood
- ▶ Two-layer MDP: denoise inner + env outer
- ▶ Hybrid: 1 SDE step / env · rest ODE $\rightarrow \sim 2\times$
- ▶ Default efficiency pick (paper preference)
- ▶ Often wins CALVIN $\pi_{0.5}$ Len-5 (87.0)

Shared recipe

few-shot SFT $\rightarrow$ freeze VLM $\rightarrow$ PPO/GAE on $\sim 300M$ action expert · chunk reward · RLinf · $8 \times H100$

Control note: chunk length H = credit-assignment knob · not classical planner

OOD honesty · what transfers

TRANSFERS

- ▶ ManiSkill visual / layout / exec shifts
- ▶ CALVIN ABC→D scene transfer
- ▶ Partial relative OOD gains ($\sim +100\%$)
- ▶ Preserves SFT OOD skills (less forgetting)

DOES NOT

- ▶ MetaWorld ML45 novel tasks — oscillates
- ▶ Cross-task skill invention via RL
- ▶ Open-world discovery narrative
- ▶ Visual OOD ceiling (frozen VLM)

Reading for Reviewer + Empiricist

RL = action-level refinement on seen skills

Not a substitute for multi-task demos or stage structure

BEHAVIOR implication: recovery / temporal efficiency — not 50-task invention

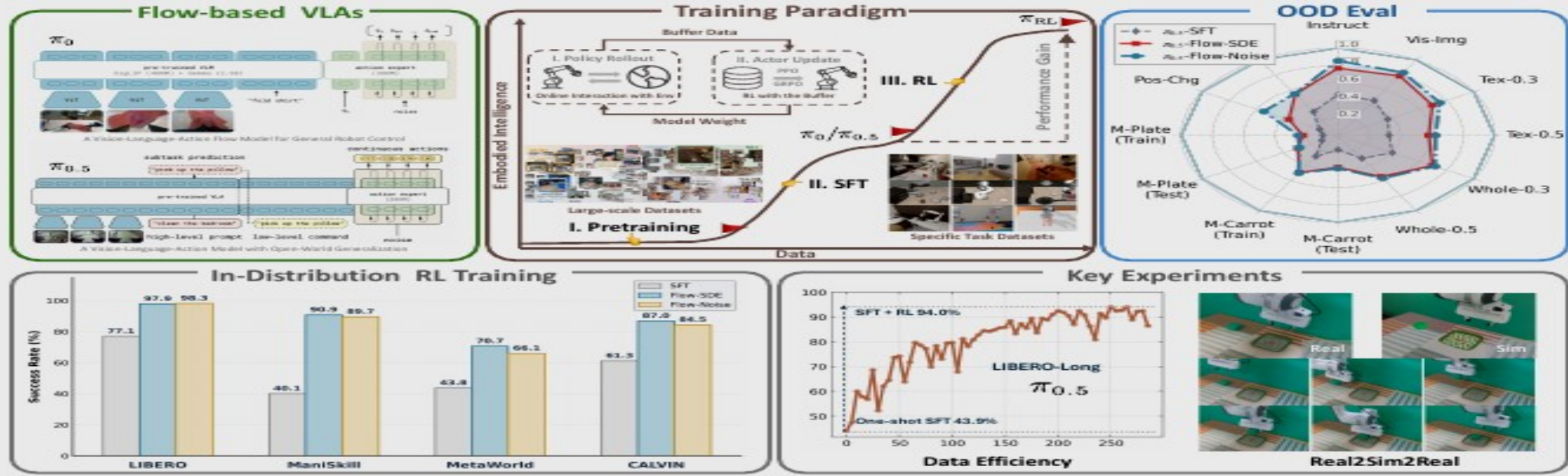


Figure 1. π_{RL} : An online RL framework for flow-based VLAs. Incorporating two solutions, Flow-Noise and Flow-SDE, π_{RL} enhances the performance and generalization of SFT-aligned models across extensive ID benchmarks and OOD settings. Refined with RL, few-shot SFT policies achieve performance comparable to full dataset baselines. Additionally, we facilitate seamless zero-shot sim-to-real transfer by constructing a simulator with 3D Gaussian Splatting as the rendering engine to narrow the visual domain gap.

Abstract

Vision-Language-Action (VLA) models enable robots to understand and perform complex tasks from multimodal input. Although recent work explores using reinforcement learning (RL) to automate the laborious data collection process in scaling supervised fine-tuning (SFT), applying RL to large-scale flow-based VLAs (*e.g.*, π_0 ,

$\pi_{0.5}$) remains challenging due to intractable action log-likelihoods raised from flow matching. We address this challenge with π_{RL} , featuring two technical approaches: (1) **Flow-Noise** models the denoising process as a discrete-time MDP with a learnable noise network for exact log-likelihood computation. (2) **Flow-SDE** integrates denoising with agent-environment interaction, formulating a two-layer MDP that employs ODE-to-SDE conversion for efficient RL exploration. We evaluate π_{RL} across various benchmarks, with experiments demonstrating that RL yields significant performance improvements in both in-distribution and out-of-distribution settings.

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Empiricist close

**Parallel sparse-success sim → specialists;
else stay on RFT. Defer Tab.1.**

π_{RL} · arXiv 2510.25889 · Oct 5 · Q&A

Unlock PPO on flow VLAs · BEHAVIOR prefers RFT

π_{RL} : Online RL Fine-tuning for Flow-based Vision-Language-Action Models

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<https://github.com/RLinf/RLinf> <https://huggingface.co/RLinf>



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